

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE CODE: CSA5113

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS: Total Marks: 50 Annexure A

Code of Conduct:

I DINDU PUJITHA(192373037) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:D PUJITHA

Annexure A

Analytical Problem Solving

1. RSA Encryption Analysis

Given $p=11$, $q=13$

- a. Compute n and $\phi(n)$
- b. Choose $e=7$ and find the private key d
- c. Encrypt the message $M=9$
- d. Decrypt the ciphertext and verify the result
- e. Analyze how changing e affects security

Answer:

Given:

- $p = 11$
- $q = 13$
- $e = 7$
- Message $M = 9$

a. Compute n and $\phi(n)$:

Step 1: Calculate n :

$$n = p \times q = 11 \times 13 = 143$$

Step 2: Calculate Euler's Totient Function:

$$\begin{aligned}\phi(n) &= (p - 1)(q - 1) \\ &= (11 - 1)(13 - 1) \\ &= 10 \times 12 = 120\end{aligned}$$

Answer

- $n = 143$
- $\phi(n) = 120$

b. Choose $e = 7$ and find the private key d :

Find d such that

$$d \times e \equiv 1 \pmod{120}$$

$$7d \equiv 1 \pmod{120}$$

Trying values,

$$7 \times 103 = 721$$

$$721 \div 120 = 6 \text{ remainder } 1$$

Hence,

$$d = 103$$

Answer

- Public Key = (7,143)
- Private Key = (103,143)

c. Encrypt the message $M = 9$:

Encryption formula

$$C = M^e \pmod{n}$$

$$C = 9^7 \pmod{143}$$

Using modular arithmetic,

$$9^2 = 81$$

$$9^4 = 81^2 = 6561 \equiv 126 \pmod{143}$$

$$9^7 = 9^4 \times 9^2 \times 9$$

$$\begin{aligned}
 &= 126 \times 81 \times 9 \\
 126 \times 81 &= 10206 \\
 10206 &\equiv 53 \pmod{143} \\
 53 \times 9 &= 477 \\
 477 &\equiv 48 \pmod{143}
 \end{aligned}$$

Ciphertext

$$C=48$$

d. Decrypt the ciphertext:

Decryption formula

$$\begin{aligned}
 M &= C^d \pmod{n} \\
 M &= 48^{103} \pmod{143}
 \end{aligned}$$

Using modular exponentiation,

$$48^{103} \pmod{143} = 9$$

Recovered Message

$$M = 9$$

The decrypted message is the same as the original message.

e. Analyze how changing *e* affects security

- Smaller values of *e* make encryption faster.
- Very small values (such as 3) may introduce security risks if padding is not used.
- Larger values increase encryption time.
- The commonly used value is **65537**, which provides a good balance between security and efficiency.

2. Block Cipher Mode Evaluation

Given plaintext blocks:

$P = [1010, 1010, 1111, 1010]$, Key = $K = 1100$

- Encrypt using ECB and CBC modes (Assume IV = 0000)
- Compare the ciphertext outputs
- Analyze which mode hides patterns better and explain why

Answer:

Given:

Plaintext Blocks: $P = 1010, 1010, 1111, 1010$

Key: $K = 1100$

IV: 0000

Assume encryption uses XOR with the key.

a. Encrypt using ECB:

Encryption $C = P \oplus K$

Block 1:

$1010 \text{ XOR } 1100 = 0110$

Block 2:

$1010 \text{ XOR } 1100 = 0110$

Block 3:

$$1111 \text{ XOR } 1100 = 0011$$

Block 4:

$$1010 \text{ XOR } 1100 = 0110$$

ECB Ciphertext:

0110, 0110, 0011, 0110

CBC Mode:**Block 1:**

$$1010 \oplus 0000 = 1010$$

$$1010 \oplus 1100 = 0110$$

$$C1=0110$$

Block 2:

$$1010 \oplus 0110 = 1100$$

$$1100 \oplus 1100 = 0000$$

$$C2=0000$$

Block 3

$$1111 \oplus 0000 = 1111$$

$$1111 \oplus 1100 = 0011$$

$$C3=0011$$

Block 4

$$1010 \oplus 0011 = 1001$$

$$1001 \oplus 1100 = 0101$$

C4=0101

CBC Ciphertext: 0110, 0000, 0011, 0101

b. Compare ciphertext outputs:

ECB: 0110
0110
0011
0110

Repeated plaintext blocks produce repeated ciphertext blocks.

CBC: 0110
0000
0011
0101

Repeated plaintext blocks produce different ciphertext blocks.

c. Which mode hides patterns better?

CBC mode hides patterns better because each plaintext block is XORed with the previous ciphertext block before encryption. Therefore, identical plaintext blocks produce different ciphertext blocks. ECB does not provide this protection.

3. DES vs 3-DES vs Blowfish Performance Study

Given:

- a. DES key size = 56 bits, time = 2 ms/block
- b. 3-DES key size = 168 bits, time = 6 ms/block
- c. Blowfish key size = 128 bits, time = 1 ms/block

For 1000 blocks of data:

- i. Calculate total encryption time for each algorithm
- ii. Analyze trade-offs between security and performance
- iii. Recommend the best algorithm for secure real-time communication

Answer:

Given:

Algorithm	Time/Block
DES	2 ms
3-DES	6 ms
Blowfish	1 ms

i. Total encryption time:

DES:

$$1000 * 2 = 2000 \text{ ms}$$

2 seconds

3-DES:

$1000 * 6 = 6000 \text{ ms}$
6 seconds

Blowfish:

$1000 * 1 = 1000 \text{ ms}$
1 second

Final Table:

Algorithm	Key Size	Total Time
DES	56 bits	2000 ms(2s)
3-DES	168 bits	6000 ms(6s)
Bloowfish	128 bits	1000 ms(1s)

ii. Security vs Performance:

DES:

- Fast
- Weak security due to short 56-bit key

3-DES:

- Stronger than DES
- Very slow

Blowfish:

- Strong security
- Fastest among the three

iii. Recommendation

Blowfish is recommended because it provides strong security with the fastest encryption speed, making it suitable for secure real-time communication

4. Diffie-Hellman Key Exchange Analysis

Given:

- a. $p=23$, $g=5$
- b. User A private key = 6
- c. User B private key = 15
- d. Compute public keys for A and B
- e. Compute the shared secret key
- f. Demonstrate how a Man-in-the-Middle attacker could alter the exchange
- g. Suggest a secure method to prevent the attack

Answer:

Given

- $p = 23$
- $g = 5$
- A's private key = 6
- B's private key = 15

d. Compute public keys:

User A: $A = g^a \bmod p = 5^6 \bmod 23$
 $5^6 = 15625$

Public key of A: 8

User B: $B = 5^{15} \bmod 23 = 19$

Public key of B: 19

e. Compute shared secret:

A computes: $19^6 \bmod 23$

Result: 2

B computes: $8^{15} \bmod 23$

Result: 2

Shared secret key: 2

f. Man-in-the-Middle Attack:

1. An attacker intercepts the exchanged public keys and replaces them with their own public keys. As a result:
2. A establishes a secret key with the attacker.
3. B also establishes a different secret key with the attacker.
4. The attacker decrypts, reads, modifies, and re-encrypts all communication without A or B realizing it.

g. Prevention:

The attack can be prevented using:

- Digital Signatures
- Digital Certificates (PKI)
- Authenticated Diffie-Hellman
- TLS/SSL protocols

These methods verify the identities of communicating parties before establishing the shared secret.

5. ECC vs RSA Comparison

Given:

- a. RSA key size = 1024 bits
- b. ECC key size = 160 bits (equivalent security)
- c. Device processing capacity = limited (IoT device)
- d. Analyze computational cost for both systems
- e. Estimate which consumes less power and memory
- f. Justify which cryptosystem is better for the given scenario with reasoning

Answer:

Given:

- RSA Key = 1024 bits
- ECC Key = 160 bits
- IoT device with limited processing power

d. Computational Cost:

RSA:

- Large key size
- More CPU computations
- Slower encryption and decryption

ECC:

- Small key size
- Faster mathematical operations
- Lower computational cost

e. Power and Memory Consumption:

RSA:

- Higher memory usage
- Higher power consumption
- Not ideal for battery-powered devices

ECC:

- Lower memory usage
- Lower power consumption
- Suitable for IoT devices

f. Which cryptosystem is better?

ECC is the better choice:

Reasons:

- Provides security equivalent to RSA with much smaller keys.
- Requires less computation.
- Consumes less memory.
- Uses less battery power.
- Faster execution on low-power devices.
- Well suited for IoT and embedded systems.